

HTML (HyperText Markup Language)

HTML is the **foundation of every website**. It is not a programming language but a **markup language** that tells the browser how to **structure and display content** such as text, images, links, and multimedia.

Modern websites combine HTML with **CSS** (styling) and **JavaScript** (functionality) to create **interactive and visually rich web applications**.

HTML Basics

Full Form:

- **HyperText** → Text that contains links to other resources or documents (hyperlinks).
- **Markup** → Tags used to **mark up** or structure content.
- **Language** → Like any language, HTML has **syntax, rules, and structure**.

Key Features of HTML:

- Provides **structure** to web pages.
 - Works with **CSS** for styling and **JavaScript** for interactivity.
 - **Platform-independent** (runs on any browser).
 - Supports **multimedia** (images, videos, audio, etc.).
 - Acts as the **first step** in web development.
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HTML Document Flow:

A typical HTML page has three essential parts:

1. **HTML Declaration** – `<!DOCTYPE html>` → Defines HTML5.

2. **Head Section** – `<head>` → Contains metadata, title, styles, and scripts (not directly visible).
 3. **Body Section** – `<body>` → Holds all the **visible content** (headings, paragraphs, images, etc.).
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HTML Syntax and Documentation

Syntax of HTML:

- HTML is made of **elements enclosed in tags**.
- Tags are written inside **angle brackets** `< >`.

Most tags come in pairs:

`<opening tag> Content </closing tag>`

Example:

`<p>Hello, this is a paragraph.</p>`

HTML Document Structure (Standard Template):

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is a basic HTML page.</p>
</body>
</html>
```

Explanation:

- `<!DOCTYPE html>` → Declare the document as HTML5.
 - `<html>` → Root element.
 - `<head>` → Metadata (title, links, scripts).
 - `<title>` → Text displayed in browser tab.
 - `<body>` → Visible content.
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HTML Tags and Elements

Tags:

- Tags are **keywords** inside angle brackets.
- Example: `<p>`, `<h1>`, `<a>`, `<img>`.

Elements:

- An element = **Opening tag + Content + Closing tag**.

Example:

```
<h2>This is a heading</h2>
```

Types of Elements:

Block-level Elements

- Occupy the full **width** of the page.
- Always **start from a new line**.
- Examples: `<div>`, `<p>`, `<h1>` to `<h6>`, `<section>`.

Inline Elements

- Occupy **only as much space as needed**.
 - Do **not** start from a new line.
 - Examples: `<span>`, `<a>`, `<img>`, `<strong>`.
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Common HTML Tags:

Tag	Purpose	Example	Type
<code><h1> ... <h6></code>	Headings (6 levels, <code><h1></code> = highest)	<code><h1>Main Heading</h1></code>	Block-level
<code><p></code>	Paragraph text	<code><p>This is a paragraph.</p></code>	Block-level
<code><a></code>	Hyperlink (link to another page/resource)	<code>Google</code>	Inline
<code></code>	Image insertion	<code></code>	Inline
<code></code>	Unordered list	<code>Item</code>	Block-level
<code></code>	Ordered list	<code>Item</code>	Block-level
<code></code>	List item	<code>Apple</code>	Block-level
<code><div></code>	Generic container (block)	<code><div>Content</div></code>	Block-level
<code></code>	Generic inline container	<code> Red Text</code>	Inline
<code><header></code>	Top section, usually contains nav/logo	<code><header><h1>Site Name</h1></header></code>	Block-level
<code><nav></code>	Navigation menu	<code><nav>Home</nav></code>	Block-level

<code><main></code>	Main content area	<code><main><p>Main article here</p></main></code>	Block-level
<code><section></code>	Group related content	<code><section><h2>About</h2><p>Details...</p></section></code>	Block-level
<code><article></code>	Independent content (blog/news)	<code><article><h2>Post</h2><p>Content...</p></article></code>	Block-level
<code><aside></code>	Side content (ads, extra info)	<code><aside><p>Related links</p></aside></code>	Block-level
<code><footer></code>	Bottom section (copyright, contact)	<code><footer>&copy; 2025 MySite</footer></code>	Block-level
<code><figure></code>	Media with caption	<code><figure><figcaption>Caption</figcaption></figure></code>	Block-level
<code><figcaption></code>	Caption for <code><figure></code>	<code><figcaption>Image Caption</figcaption></code>	Block-level
<code><mark></code>	Highlight text	<code><p>This is <mark>important</mark>.</p></code>	Inline
<code><time></code>	Represent date/time	<code><time datetime="2025-10-03">Oct 3, 2025</time></code>	Inline
<code></code>	Important text (bold)	<code>Important!</code>	Inline
<code></code>	Emphasized text (italic)	<code>Emphasized</code>	Inline
<code>
</code>	Line break	<code>Line1
Line2</code>	Inline

<code><code></code>	Inline code snippet	<code><code>console.log('Hello')</code></code>	Inline
<code><blockquote></code> <code>></code>	Block quote (long text quote)	<code><blockquote>Quote here</blockquote></code>	Block-level
<code><pre></code>	Preformatted text	<code><pre>Text aligned</pre></code>	Block-level
<code><small></code>	Smaller text	<code><small>Note</small></code>	Inline

Semantic HTML

What is Semantic HTML?

- **Semantic HTML** uses tags that **describe the meaning** of the content, instead of just appearance.
- Example:
 - Non-semantic: `<div id="header"> ... </div>`
 - Semantic: `<header> ... </header>`

Benefits of Semantic HTML:

- **SEO Friendly** → Helps search engines understand structure.
- **Accessibility** → Screen readers and assistive tools interpret content better.
- **Readability** → Developers instantly recognize content purpose.

Common Semantic Elements:

Semantic Tag	Description	Example
<code><header></code>	Top section, usually contains logo, nav	<code><header><h1>Site Name</h1></header></code>
<code><nav></code>	Navigation bar (menus, links)	<code><nav>Home</nav></code>

<code><main></code>	Main content of the page	<code><main><p>Main article here</p></main></code>
<code><section></code>	Groups related content	<code><section><h2>About</h2><p>Details...</p></section></code>
<code><article></code>	Independent content (blog/news posts)	<code><article><h2>Post</h2><p>Blog content...</p></article></code>
<code><aside></code>	Side content (ads, sidebars, extra info)	<code><aside><p>Related links</p></aside></code>
<code><footer></code>	Bottom section (copyright, contact)	<code><footer>&copy; 2025 MySite</footer></code>
<code><figure> + <figcaption></code>	Image with caption	<code><figure><figcaption>Image Caption</figcaption></figure></code>
<code><time></code>	Represents date/time	<code><time datetime="2025-10-03">Oct 3, 2025</time></code>

Example of Semantic HTML in Action:

```

<!DOCTYPE html>
<html>
<head>
  <title>Semantic HTML Example</title>
</head>
<body>
  <header>
    <h1>My Website</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Blog</a>
    </nav>
  </header>

  <main>

```

```
<section>
  <h2>About Me</h2>
  <p>I am a web developer who loves coding.</p>
</section>
<article>
  <h2>My First Blog</h2>
  <p>This is my first blog article written using semantic HTML.</p>
</article>
</main>

<aside>
  <h3>Quick Links</h3>
  <p>Visit my <a href="#">Portfolio</a></p>
</aside>

<footer>
  <p>&copy; 2025 My Website. All rights reserved.</p></footer>
</body>
</html>
```